

Cell Cryopreservation Market Size, Regional Trends, Growth Forecast - 2036

The cell cryopreservation market was valued at USD 14.32 billion in 2025 and is expected to reach USD 101.98 billion by 2036, registering a CAGR of 18.91% during the forecast period (2026-2036). Market growth is fueled by the increasing use of regenerative medicine, expansion of cell and gene therapies, rising demand for biobanking services, and continuous investment in biomedical research. Advances in cryogenic storage technologies and growing applications of preserved cells in clinical and research settings are expected to further accelerate market development.

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Cell Cryopreservation Industry Demand

The [cell cryopreservation market](#) comprises products and technologies used to preserve living cells at extremely low temperatures while maintaining their biological integrity for future use. Cryopreservation is an essential process in stem cell research, regenerative medicine, fertility treatments, drug development, tissue engineering, and clinical diagnostics. It enables long-term storage without significantly affecting cell viability, making it indispensable across modern healthcare and life sciences.

Demand for cell cryopreservation solutions continues to rise due to the growing need for reliable storage of biological samples, increasing adoption of personalized medicine, and expansion of cell-based therapies. The market also benefits from the cost-effectiveness of preserving valuable biological materials compared to repeated sample collection. Standardized preservation protocols simplify handling, while extended shelf life supports long-term storage and global transportation. Increasing investments in biotechnology research, biobanks, and pharmaceutical manufacturing are further strengthening industry demand.

Cell Cryopreservation Market: Growth Drivers & Key Restraint

Growth Drivers

- **Expansion of Cell and Gene Therapy:** The rapid development of cell- and gene-based therapies has significantly increased the demand for advanced cryopreservation technologies. Safe preservation ensures that therapeutic cells remain viable throughout manufacturing, storage, transportation, and clinical use.
- **Growing Outsourcing Activities:** Biopharmaceutical companies are increasingly outsourcing storage and manufacturing operations to CMOs and CDMOs. These partnerships reduce operational costs, improve production efficiency, and provide access to advanced cryogenic storage infrastructure.

- **Rising Research and Biobanking Investments:** Governments, research institutes, and pharmaceutical companies are investing heavily in stem cell research, precision medicine, and disease-specific biobanks. This has created strong demand for reliable cryopreservation media, equipment, and consumables.

Restraint

- **High Operational Costs:** Advanced cryopreservation systems require specialized equipment, continuous temperature monitoring, skilled personnel, and regulatory compliance. These requirements increase operational expenses and can limit adoption among smaller laboratories and healthcare facilities.

Cell Cryopreservation Market: Segment Analysis

Segment Analysis by Application

Stem Cell Banking

Stem cell banking remains one of the leading applications due to growing awareness of regenerative medicine and future therapeutic use. Increasing public and private banking initiatives continue to support strong market demand.

Assisted Reproductive Technology

Cryopreservation is widely used for preserving embryos, eggs, and sperm during fertility treatments. Rising infertility rates and growing acceptance of assisted reproductive procedures are driving continuous expansion of this segment.

Cell Therapy

The increasing commercialization of cell therapies has strengthened demand for high-quality preservation solutions. Reliable cryopreservation supports efficient storage and transportation while maintaining treatment effectiveness.

Drug Discovery and Development

Pharmaceutical and biotechnology companies depend on preserved cell lines for drug screening, toxicity testing, and biomedical research. Continuous research activities keep this application segment highly attractive.

Others

Additional applications include vaccine research, tissue engineering, academic research, and disease modeling, where long-term preservation of viable cells remains essential.

Segment Analysis by End Use

Biopharmaceutical & Pharmaceutical Companies

These organizations represent a major share of market demand due to continuous research, biologics manufacturing, and development of advanced therapies requiring reliable cryogenic storage.

CMOs and CDMOs

Contract service providers continue to experience rising demand as pharmaceutical companies outsource manufacturing and preservation activities to improve efficiency and scalability.

Biobanks

Biobanks are expanding rapidly to support clinical research, precision medicine, and population health studies. Growing collections of biological samples continue to drive demand for advanced cryopreservation systems.

IVF Clinics

Fertility clinics increasingly rely on cryopreservation for reproductive cells and embryos, supporting improved treatment flexibility and successful reproductive outcomes.

Others

Hospitals, academic institutions, research laboratories, and diagnostic centers utilize cryopreservation technologies for clinical research and specialized laboratory applications.

Segment Analysis by Product

Cell Freezing Media

Cell freezing media remain essential for protecting cells during freezing and thawing, ensuring high post-thaw viability across research and therapeutic applications.

Ethylene Glycol

Ethylene glycol is widely used as a cryoprotectant for selected biological samples because it minimizes cellular damage during freezing.

Dimethyl Sulfoxide

Dimethyl sulfoxide remains one of the most commonly used cryoprotective agents due to its excellent ability to preserve cell integrity during long-term storage.

Glycerol

Glycerol is extensively utilized for preserving blood cells and microbial cultures, making it an important component of cryopreservation protocols.

Others

Alternative cryoprotective formulations are being developed to improve preservation efficiency while reducing cellular toxicity.

Equipment

Cryopreservation equipment ensures controlled freezing, secure storage, and continuous monitoring, forming the foundation of modern cryogenic laboratories.

Freezers

Ultra-low temperature freezers are widely used across hospitals, research institutions, and pharmaceutical facilities for reliable biological sample storage.

Incubators

Incubators support controlled laboratory conditions during sample preparation and recovery before and after cryopreservation.

Liquid Nitrogen Supply Tanks

Liquid nitrogen tanks are indispensable for maintaining ultra-low temperatures required for long-term preservation of biological materials.

Others

Additional storage and monitoring systems improve operational efficiency, sample tracking, and regulatory compliance.

Consumables

Consumables play an important role in safe storage, labeling, transportation, and handling of cryopreserved samples.

Cryogenic Vials

Cryogenic vials provide durable, leak-resistant storage for valuable biological specimens under ultra-low temperature conditions.

Cryogenic Tubes

Cryogenic tubes support secure preservation and organized sample management in research and clinical laboratories.

Cooler Boxes/Containers

Specialized transport containers help maintain required temperatures during the shipment of biological samples.

Others

Additional accessories, including storage racks and labeling solutions, enhance laboratory workflow and sample organization.

Cell Cryopreservation Market: Regional Insights

North America

North America leads the market due to its advanced biotechnology industry, strong pharmaceutical research ecosystem, extensive biobanking infrastructure, and growing adoption of regenerative medicine. Continuous innovation and supportive regulatory frameworks sustain regional demand.

Europe

Europe maintains a significant market position through increasing investments in stem cell research, expanding clinical trials, and well-established healthcare infrastructure. Strong academic collaborations and supportive regulatory policies continue to encourage market growth.

Asia-Pacific (APAC)

Asia-Pacific is experiencing rapid market expansion owing to improving healthcare infrastructure, growing biotechnology investments, increasing fertility treatment procedures, and rising government support for life science research. The expansion of pharmaceutical manufacturing and biobanking facilities is further strengthening regional demand.

Top Players in the Cell Cryopreservation Market

The cell cryopreservation market is characterized by continuous innovation and strategic collaborations among leading industry participants. Key companies include Thermo Fisher Scientific Inc. (U.S.), Miltenyi Biotec GmbH (Germany), BioLife Solutions Inc. (U.S.), PromoCell GmbH (Germany), Corning Incorporated (U.S.), Cooper Companies (U.S.), Bio-Rad Laboratories (India), Creative Biolabs (U.S.), Lonza (Switzerland), and HiMedia Laboratories (India). These organizations focus on expanding product portfolios, enhancing cryopreservation technologies, developing advanced storage solutions, and strengthening global distribution networks to meet the growing demand for regenerative medicine, cell therapy, and biobanking applications.

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